

RAYWIN CRUZ THUMMA RAYAPPAN

Working Student – AI & Generative AI Prototyping (SAP HANA Cloud)

M.Sc. Data Science – University of Mannheim

PERSONAL DETAILS

Location:	Mannheim, Germany
Mobile:	+49 177 3656605
Email:	raywincruz07@gmail.com
LinkedIn:	www.linkedin.com/in/raywincruz
Work Authorization:	Valid Student residence permit (up to 20h/week)

PROFILE

M.Sc. Data Science student at the University of Mannheim specializing in **Generative AI, LLM architectures (RAG-based systems), and business-oriented AI prototyping**. Experienced in building **end-to-end ML pipelines, structured data validation systems, and translating technical AI capabilities** into customer-facing use cases. Passionate about accelerating enterprise **AI adoption within SAP HANA Cloud Product Management**.

EDUCATION

09/2025 – Present	MSc in Data Science - University of Mannheim Mannheim, Germany Expected graduation: 09/2027 • Relevant Focus Areas: Machine Learning & Generative AI, Data Mining & Business Data Analysis, Knowledge Graphs & Structured Information Processing, AI Product Development
08/2021 – 06/2025	BE in Computer Science - Sathyabama University Chennai, India • Strong foundation in structured analysis, documentation, and project coordination • Bachelor Project: AI-Based Logistics Management System • Designed and prototyped an AI-driven decision-support system for logistics optimization • Built structured data pipelines (raw → validated → analytics-ready) • Implemented ML-assisted rule-based optimization logic • Published research paper in a peer-reviewed journal

PRACTICAL EXPERIENCE

07/2024 – 07/2025	Data Analysis – Internship Martex AI – Chennai, India • Designed Python-based analytical pipelines for enterprise data processing workflows • Built structured validation and consistency frameworks to ensure high-quality datasets for AI systems • Translated business requirements into technical prototypes and automation tooling • Reduced data inconsistencies through automated validation framework • Supported end-to-end lifecycle from raw ingestion to production-ready outputs • Collaborated cross-functionally with analytics and engineering teams
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04/2024 – 05/2024	Analytics Internship – Payment Fraud Detection EDUTECH LIFE – Gurugram, India • Supported ML-based fraud detection workflows using Python and SQL • Assisted in feature validation and anomaly detection logic for transactional datasets • Assisted in translating analytical findings into structured reports for decision-making teams • Documented model performance insights for stakeholder discussions • Contributed to operational evaluation of AI use cases under real-world constraints
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PROJECTS

09/2025 – 01/2026	Data Mining End-to-End ML & AI Prototyping Project • Designed and implemented a complete ML pipeline in Python • Performed feature engineering and model evaluation for structured business data • Translated model outputs into explainable decision-support insights • Documented findings and created presentation-ready summaries for stakeholders
09/2025 – 01/2026	Web Data Integration Project • Designed heterogeneous data ingestion pipelines across multiple sources • Implemented validation rules, consistency checks, and quality assurance logic • Created AI-ready structured datasets optimized for downstream ML workflows • Maintained detailed technical documentation for reproducibility and collaboration

SKILLS

AI & Machine Learning:	Generative AI Concepts, LLM-based Systems (RAG understanding) Feature Engineering, Model Evaluation, Structured Data Processing
Programming:	Python (ML pipelines, automation, data engineering) SQL (advanced queries, joins, optimization) JavaScript (basic)
Data & Systems:	Relational Databases (RDBMS), Data Validation & Quality Assurance Structured Pipeline Design, Log-driven debugging & metrics analysis
Cloud & Tools:	Docker (basic), Kubernetes (conceptual), Git Linux , Jupyter , Power BI (dashboarding)
LANGUAGES:	English – Fluent (written & spoken) German – A1 (actively improving)

ADDITIONAL INFORMATION

- **Availability:** 15–20 hours/week | Immediate Start
- **On-site availability** in Walldorf
- **Long-term availability** (minimum 3 semesters)
- **Research Publications:** Published a research paper on an **AI-based logistics management system** <https://iarjset.com/papers/ai-based-logistics-management-system/>